Comparative Analysis of Embedded AI Techniques for Bovine Meat Monitoring

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***Abstract*—To prevent foodborne illnesses resulting from the consumption of spoiled meat, it is essential to ensure the freshness of the meat in question. Consequently, impedance spectroscopy can be employed to achieve this objective. Nevertheless, in order to ensure the necessary portability, it is essential to incorporate sophisticated analytical techniques into the embedded system. This paper presents a comparative analysis of embedded artificial intelligence (AI) techniques for bovine meat monitoring. We implement and evaluate three machine learning algorithms-neural networks (NN), support vector machines (SVM), and decision trees on two embedded platforms: Coral and Arduino. The results demonstrate high classification accuracies, with a neural network achieving 99.47% on the Coral platform and a support vector machine reaching 99.5% on the Arduino platform when processing augmented impedance data across four meat classes based on their freshness. The trade-offs between accuracy, speed, and memory usage across different combinations of AI models and hardware platforms are discussed, providing insights for optimal embedded AI solutions in meat quality assessment.**

***Index Terms*—Embedded AI, Bovine Meat Monitoring, Neural Networks, Support Vector Machines, Decision Trees, Coral, Arduino**

I. Introduction

Ensuring the quality and safety of bovine meat is a critical issue in the global food industry, given its significant implications for public health, consumer confidence, and the economic stability of the meat sector. Meat spoilage, mainly due to bacterial growth, poses a risk of foodborne illnesses, highlighting the need for effective freshness assessment as an essential part of food safety protocols. Traditionally, the evaluation of meat freshness has relied on human sensory perception, particularly sight and smell. However, these methods are inherently subjective and can vary between different assessors. This subjectivity, combined with the growing demand for rapid and precise quality control in large-scale meat production, has driven the development of more objective and automated assessment methods.

Recently, impedance spectroscopy (IS) has gained attention as a promising alternative for automating the assessment of meat freshness. IS offers several benefits over traditional methods, including non-destructive testing, fast results, and the ability to detect subtle changes in meat composition. Numerous studies have confirmed the effectiveness of IS in evaluating meat quality [1]–[5]. Traditional meat monitoring systems rely on a variety of sensors and equipment to measure parameters such as temperature, pH, and microbial activity. These systems require precise calibration, regular maintenance, and often multiple devices to capture different data types. Some advanced methods even incorporate high-quality cameras for real-time image collection, GPUs for processing, and STEM techniques, making the systems more complex and costly [6].

To date, the majority of meat quality assessment implementations have been confined to PC platforms and reliant on cloud data storage, thereby limiting their practical application in on-site or in-line monitoring scenarios [7]. These traditional systems often involve complex, multi-sensor setups, which can be costly and challenging to implement in real-world environments. However, rapid advancements in Artificial Intelligence (AI) and embedded systems present novel opportunities for enhancing meat quality assessment [8]. In contrast to traditional methods, embedded AI systems use fewer, more versatile sensors [9], [10]. They often utilize spectroscopy or simple impedance measurements, with AI algorithms processing the data to infer multiple quality parameters. This approach significantly reduces the need for complex setups, leading to more streamlined and cost-effective solutions. Microcontrollers and single-board computers, such as the Arduino BLE Nano 33 and Coral Dev Board Mini, offer compact, energy-efficient platforms capable of executing sophisticated algorithms [11]. These developments raise intriguing questions regarding the feasibility of implementing shallow learning techniques on such devices and the potential improvements in speed and accuracy they might offer for meat quality assessment.

This paper explores the application of embedded AI techniques for bovine meat monitoring, with a specific focus on implementing various machine learning algorithms on resource-constrained devices. The primary objective is to evaluate and compare the performance of conventional shallow learning models, i.e., neural networks (NN), support vector machines (SVM), and decision trees, on two popular embedded platforms: Coral Dev Board Mini and Arduino BLE Nano 33. These combinations are evaluated based on three critical metrics: classification accuracy, processing speed, and memory usage [12].

II. Methodology

A. Data Collection and Preprocessing

Following impedance spectroscopy measurement, the data are processed using two complementary methods: the Equivalent Circuit Model (ECM) and the extraction of characteristic

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2024 International Workshop on Impedance Spectroscopy (IWIS) | 979-8-3315-0762-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/IWIS63047.2024.10847102

points. These methods allow for a comprehensive analysis of the impedance data, yielding a total of 7 features that characterize the meat sample's electrical properties.

To develop, the ECM approach involves fitting the impedance data to an electrical circuit model that represents the physical and chemical properties of the meat sample [13]. This model typically consists of resistors and capacitors arranged in a specific configuration to mimic the electrical behavior of biological tissues.

The ECM fitting process yields several features:

a) R_e: Represents the resistance of the extracellular fluid

b) R_i: Indicates the resistance of the intracellular fluid

c) C_m: Reflects the capacitance of cell membranes

d) α: A dimensionless parameter related to the distribution of relaxation times

This method involves identifying and analyzing specific points of interest in the impedance spectrum. These points often correspond to critical frequencies or impedance values that are particularly sensitive to changes in meat freshness [2].

The data are collected from daily impedance spectra measurements of bovine meat. Interpolation is applied to the extracted data to augment the dataset, resulting in 1883 samples. Fig. 1 shows the class distribution of meat samples. The classified meat data are divided into 4 classes that correspond to 4 different periods of time: period 1: between day 2 and 6 (fresh meat), period 2: between day 7 and 9 (edible), period 3: between day 10 and 12 (critical), and period 4: more than day 13 (dangerous). The data are then divided into training (60%), validation (20%), and test (20%) sets.

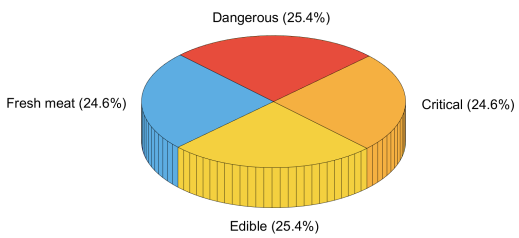

Fig. 1. Class Distribution of Meat Samples

B. Hardware Platforms

To evaluate the performance of the classification algorithms in embedded systems, two distinct hardware platforms were used to represent different ends of the spectrum in terms of computational capabilities and power consumption: the Coral Dev Board Mini (a single-board computer) and The Arduino Nano 33 BLE (a microcontroller).

1) Coral Dev Board Mini: The Coral Dev Board Mini (see Fig. 2) features a dedicated Tensor Processing Unit (TPU) capable of up to 4 TOPS for accelerated ML tasks, supported by a quad-core ARM Cortex-A35 CPU for general processing. It includes 2 GB of RAM and 8 GB of flash memory, expandable via microSD. This combination provides significant computational power for running complex AI models efficiently at the edge, making it ideal for real-time applications.

Fig. 2. Google Coral Dev Board Mini

2) Arduino BLE Nano 33: The Arduino Nano 33 BLE (see Fig. 3) features an nRF52840 SoC with a 32-bit ARM Cortex-M4 CPU, running at 64 MHz. It has 1 MB of flash memory and 256 KB of RAM. While not designed for heavy computational tasks, it excels in low-power IoT applications and can handle simple ML models, making it ideal for energy-efficient, real-time processing tasks.

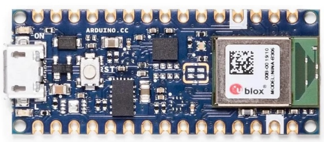

Fig. 3. Arduino Nano 33 BLE

C. Machine Learning Models

The implementation and comparison of three traditional machine learning (ML) models, each selected for their established effectiveness in classification tasks and suitability for deployment in embedded systems, are outlined below:

- Neural Network (NN): A feedforward multilayer perceptron with an architecture designed for predefined specific classification task. The model consists of:
  - Input layer: 9 neurons (corresponding to our 7 impedance features plus 2 additional contextual features)
  - Three hidden layers: 16 neurons each, with ReLU activation functions
  - Output layer: 4 neurons (representing the four freshness categories), with softmax activation

  This relatively shallow architecture was chosen for its balance between performance and computational efficiency, making it suitable for embedded systems [14].
- Support Vector Machine (SVM): Implemented using the scikit-learn library and subsequently converted into a TensorFlow Lite model for embedded deployment. The SVM utilized a radial basis function (RBF) kernel, which is

well-suited for addressing non-linear classification problems. [15], [16]. SVMs are known for their robustness in high-dimensional spaces and have performed well in various food quality assessment tasks [17].
- Decision Tree: Constructed using the "DecisionTreeClassifier" from scikit-learn, with hyperparameters optimized for meat freshness classification. Decision trees offer the advantage of interpretability and can capture non-linear relationships in the data. They are also computationally efficient, making them well-suited for resource-constrained environments.

Neural Networks, Support Vector Machines, and Decision Trees were selected for the study as they represent key approaches in machine learning, balancing computational efficiency, adaptability to complex patterns, and interpretability. These conventional models are widely applied in various classification tasks, including food quality assessment, allowing for a comprehensive evaluation of different algorithms' effectiveness [18].

III. Results and Discussion

A. Classification Accuracy

Table I shows the classification accuracies achieved by different models on both platforms. In general, all the methods have obtained similar results with an accuracy around 98%. However, due to the limitations of each board and the fact that Coral only supports neural network architecture and SVM, the SVM on Coral was less accurate than that on Arduino. In terms of accuracy, all models give similar results.

TABLE I
TEST ACCURACY COMPARISON

Model	Coral	Arduino
Neural Network	99.47%	Not supported
SVM	96.02%	99.5%
Decision Tree	Not supported	99.73%

B. Processing Speed

The processing speed encompasses both the inference time and the overhead associated with preparing the model for classification. Inference time refers to the duration required for the model to process input data and produce results, while overhead includes the time spent on tasks such as model loading and preprocessing. The measurement of execution time for each model was conducted on both platforms, incorporating these factors. Table II shows the average processing time per classification. While the classification time for Coral is approximately a few milliseconds, the overhead can be particularly time-consuming. Specifically, it was found that Coral requires approximately 5 seconds to complete the classification task. In comparison, Arduino requires less time, approximately 3.5 to 4.2 seconds, to perform the same tasks. While a direct comparison is only possible for SVM, it can be observed that Coral takes approximately 1 second longer to make a classification than Arduino.

TABLE II
PROCESSING SPEED COMPARISON IN SECONDS

Model	Coral	Arduino
Neural Network	5.28	Not supported
SVM	5.26	4.21
Decision Tree	Not supported	3.53

C. Memory Usage

Table III presents the memory consumption of different models on both platforms.

TABLE III
MEMORY USAGE COMPARISON (KB)

Model	Coral	Arduino
Neural Network	12	Not supported
SVM	8	99
Decision Tree	Not supported	83

As Coral relies on TensorFlow Lite, the models typically occupy less than 15 kB of memory. In its unmodified form and when applied to Arduino, it can be observed that the majority of models require approximately 90 kB. Consequently, the library-specific implementation can have a significant impact. TensorFlow Lite compresses the data by applying quantization, which is not available for scikit-learn models. This explains the considerable discrepancy between the different implementations.

IV. Interpretation

This study compared the performance of three conventional machine learning models-Neural Network, Support Vector Machine (SVM), and Decision Tree-on two embedded platforms, the Coral Dev Board Mini and the Arduino Nano 33 BLE, for meat freshness classification. The results revealed several key insights into the trade-offs and considerations for deploying AI-based meat quality assessment in resource-constrained environments.

Classification accuracies across all models were impressively high, ranging from 96.02% to 99.5%, underscoring the effectiveness of impedance spectroscopy-based features for meat freshness classification. This indicates that even relatively simple machine learning models can achieve high performance when provided with the right input features. However, the study also highlighted the significant impact of hardware limitations and software implementations on the feasibility of model deployment.

Platform compatibility varied notably between the two boards. The Coral Dev Board Mini excelled in Neural Network implementations but was unable to support Decision Trees. Conversely, the Arduino Nano 33 BLE performed well with SVM and Decision Trees but could not support Neural Networks due to memory constraints. Interestingly, the Arduino platform exhibited faster execution times (3.53-4.21 seconds) compared to the Coral (5.26-5.28 seconds), largely due to the overhead associated with TensorFlow Lite on the Coral board. This finding emphasizes that raw computational power alone is not the sole determinant of performance in embedded AI applications; the efficiency of the software stack also plays a crucial role.

Memory usage patterns showed that the Coral's use of TensorFlow Lite resulted in significantly smaller model sizes (8-12 kB) compared to the Arduino's unmodified implementations (83-99 kB). This highlights the importance of model compression techniques, such as quantization, in embedded deployments. Although all models achieved high accuracy, choosing the best model involves balancing factors like processing speed, memory usage, and hardware compatibility. For example, the SVM model provides a strong middle ground, being implementable on both platforms with high accuracy, albeit with different memory and speed characteristics.

V. Conclusion

The findings of this study have significant implications for the practical implementation of AI-based meat freshness classification systems. The choice between the Coral Dev Board Mini and Arduino Nano 33 BLE should be guided by specific application requirements. The Coral platform may be more suitable for applications that prioritize model complexity and future expandability, while the Arduino platform shows advantages for those that emphasize energy efficiency and faster deployment. The study also demonstrates the critical impact of software implementation on model size and performance, suggesting that further optimization, especially for the Arduino platform, could lead to substantial improvements.

Although the study primarily focused on accuracy, processing speed, and memory usage, real-world deployments must also consider factors such as power consumption, durability, and ease of integration into existing meat processing workflows. Future research should explore advanced model compression techniques to further reduce the memory footprint, particularly for the Arduino platform. Additionally, investigating the impact of varying environmental conditions and meat types on classification accuracy is essential to ensure robustness in diverse real-world settings. Developing hybrid approaches that leverage the strengths of both platforms could also be beneficial, such as using the Arduino for rapid, energy-efficient preliminary assessments and the Coral for more complex analyses when necessary.

In conclusion, this study demonstrates the feasibility and potential of embedded AI for meat freshness classification on different hardware platforms. By carefully considering trade-offs between model complexity, accuracy, processing speed, and memory usage, it is possible to develop effective, efficient, and deployable solutions for the on-site evaluation of meat quality. These findings pave the way for more widespread adoption of AI-driven quality control in the meat industry, with the potential to improve food safety, reduce waste, and enhance consumer confidence.

The findings of this study have far-reaching implications, not only for the meat industry but also for the broader field of food safety and quality assessment. By demonstrating the viability of embedded AI solutions, the authors pave the way for more widespread adoption of automated, rapid, and accurate meat quality assessment techniques, ultimately contributing to enhanced food safety standards and consumer confidence in the global meat supply chain.

Acknowledgment

The authors disclose that artificial intelligence (AI)-generated text has been used in the preparation of this manuscript, in compliance with IEEE policies [19]. Specifically, Claude (Anthropic) [20] and ChatGPT (OpenAI) [21] were utilized to assist in refining the manuscript's structure, enhancing language clarity, and improving overall coherence. All AI-generated content was thoroughly reviewed and edited by the authors, who retain full responsibility for the paper's content, analysis, and conclusions.

References

[1] M. Guermazi, O. Kanoun, and N. Derbel, "Method of model's parameters classification using neural network for meat characterization," in *10th International Multi-Conferences on Systems, Signals & Devices 2013 (SSD13)*. IEEE, 2013.

[2] J.-L. Damez, S. Clerjon, S. Abouelkaram, and J. Lepetit, "Beef meat electrical impedance spectroscopy and anisotropy sensing for non-invasive early assessment of meat ageing," *Journal of Food Engineering*, vol. 85, no. 1, pp. 116–122, 2008.

[3] X. Zhao, H. Zhuang, S.-C. Yoon, Y. Dong, W. Wang, and W. Zhao, "Electrical impedance spectroscopy for quality assessment of meat and fish: A review on basic principles, measurement methods, and recent advances," *Journal of Food Quality*, vol. 2017, no. 1, p. 6370739, 2017.

[4] M. Guermazi, O. Kanoun, and N. Derbel, "Investigation of long time beef and veal meat behavior by bioimpedance spectroscopy for meat monitoring," *IEEE Sensors Journal*, vol. 14, no. 10, pp. 3624–3630, 2014.

[5] S. Huh, H.-J. Kim, S. Lee, J. Cho, A. Jang, and J. Bae, "Utilization of electrical impedance spectroscopy and image classification for non-invasive early assessment of meat freshness," *Sensors*, vol. 21, no. 3, p. 1001, 2021.

[6] M. Nami, M. Taheri, J. Siddiqui, I. A. Deen, M. Packirisamy, and M. J. Deen, "Recent progress in intelligent packaging for seafood and meat quality monitoring," *Advanced Materials Technologies*, p. 2301347, 2024.

[7] D. Kolosov, L.-C. Fengou, J. M. Carstensen, N. Schultz, G.-J. Nychas, and I. Mporas, "Microbiological quality estimation of meat using deep cnns on embedded hardware systems," *Sensors*, vol. 23, no. 8, pp. 1234–1248, 2023.

[8] Y. Shi, X. Wang, M. S. Borhan, J. Young, D. Newman, E. Berg, and X. Sun, "A review on meat quality evaluation methods based on non-destructive computer vision and artificial intelligence technologies," *Food science of animal resources*, vol. 41, no. 4, p. 563, 2021.

[9] M. Mnif, S. Sahnoun, M. Kaaniche, B. B. Atitallah, A. Fakhfakh, and O. Kanoun, "Ultra-fast edge computing approach for hand gesture classification based on eit measurements," in *2024 IEEE International Symposium on Robotic and Sensors Environments (ROSE)*. IEEE, 2024, pp. 1–7.

[10] M. Mnif, S. Sahnoun, M. Djemaa, A. Fakhfakh, and O. Kanoun, "Exploring model compression techniques for efficient 1d cnn-based hand gesture recognition on resource-constrained edge devices," in *2024 IEEE 7th International Conference on Advanced Technologies, Signal and Image Processing (ATSIP)*, vol. 1. IEEE, 2024, pp. 659–664.

[11] S. C. Magalhães, F. N. dos Santos, P. Machado, A. P. Moreira, and J. Dias, "Benchmarking edge computing devices for grape bunches and trunks detection using accelerated object detection single shot multibox deep learning models," *Engineering Applications of Artificial Intelligence*, vol. 117, p. 105604, 2023.

[12] S. GC *et al.*, "Using deep learning neural network in artificial intelligence technology to classify beef cuts," *Frontiers in Sensors*, vol. 2, p. 654357, 2021.

[13] U. Pliquett, "Bioimpedance: a review for food processing," *Food engineering reviews*, vol. 2, pp. 74–94, 2010.

[14] L. Pan, W. Zhang, N. Zhu, S. Mao, and K. Tu, "Early detection and classification of pathogenic fungal disease in post-harvest strawberry fruit by electronic nose and gas chromatography–mass spectrometry," *Food Research International*, vol. 62, pp. 162–168, 2014.
[15] A. Mammone, M. Turchi, and N. Cristianini, "Support vector machines," *Wiley Interdisciplinary Reviews: Computational Statistics*, vol. 1, no. 3, pp. 283–289, 2009.
[16] C. Cortes and V. Vapnik, "Support-vector networks," *Machine learning*, vol. 20, pp. 273–297, 1995.
[17] S. R. Safavian and D. Landgrebe, "A survey of decision tree classifier methodology," *IEEE transactions on systems, man, and cybernetics*, vol. 21, no. 3, pp. 660–674, 1991.
[18] N. G. Baltas, P. Mazidi, J. Ma, F. de Asis Fernandez, and P. Rodriguez, "A comparative analysis of decision trees, support vector machines and artificial neural networks for on-line transient stability assessment," in *2018 International Conference on Smart Energy Systems and Technologies (SEST)*. IEEE, 2018, pp. 1–6.
[19] IEEE, "Submission policies," https://conferences.ieeeauthorcenter.ieee.org/author-ethics/guidelines-and-policies/submission-policies/, 2024, accessed: 2024-09-03.
[20] Anthropic, "Claude," https://www.anthropic.com, 2024, accessed: 2024-09-03.
[21] OpenAI, "ChatGPT," https://www.openai.com/chatgpt, 2024, accessed: 2024-09-03.